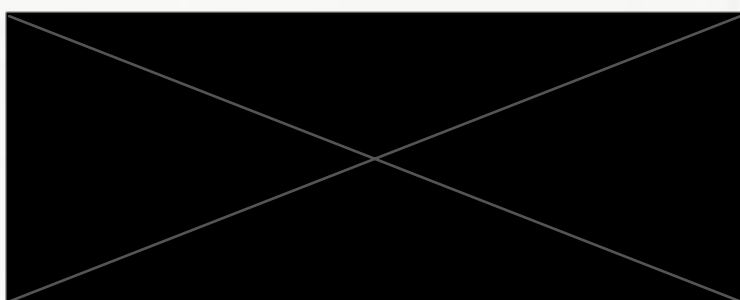




30 JULY 1985

Dear 

Thanks immensely for sending me a copy of your paper. I think it is quite fundamental, and you are to be highly commended for your insight and development. Also it was a great pleasure to finally get to meet you; I'm only sorry that I didn't get to talk to you at greater length.

This year the conference was tremendous. Pharis Williams' work is absolutely fundamental, as was the work Henry Monteth presented on a fundamental glitch in present EM. Rauscher's work is very important, and I was delighted to meet Fred Blau and hear his presentations. Several other very good papers were presented.

I'm enclosing a couple of things in case you do not have copies. (Any papers of mine that are of any use to you, I will be glad to send you).

It is increasingly apparent that the vector analysis axioms for the zero vector,

$$\vec{v} + \vec{0} = \vec{0} + \vec{v} \quad (1)$$

really did it to us. Put subscripts on, such as

$$\vec{v} + \vec{0}_i = \vec{0}_j + \vec{v} \quad (2)$$

and it becomes obvious that this axiom

(1) makes all zero-vectors equal, and equal to zero, and (2) negates any differentiation between systems of vectors that sum to a zero resultant. In fact, the conventional practice presently is to replace such a system with a zero vector.*

However, the QM view is that, at fundamental level, all force fields are effects produced by an operator. Thus, in a QM view, if we have

$$\vec{A} + \vec{B} + \vec{C} = \vec{0} \quad (3)$$

then the operators for $\vec{A}$, $\vec{B}$, and $\vec{C}$ may still exist, even though the resultant of all the operators is a zero vector. Thus such a system can be a dynamic thing, for example, if $\vec{A} = \vec{A}(t)$, $\vec{B} = \vec{B}(t)$, $\vec{C} = \vec{C}(t)$.

* For dynamic ^{zero} systems of time-varying fields related with a potential.

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In this respect, another such system,

$$\bar{D} + \bar{E} + \bar{F} = \bar{0} \quad (4)$$

may be very different indeed from the first system.

Further, after a session with Pharis Williams, he pointed out that in general relativity, we would have the energy density in vacuum of an EM wave to be a function of $E^2 + B^2$, and for a wave 180° out, $(-E)^2 + (-B)^2$ is still positive. That is, the energy densities add, even though the force fields have vanished.

My addition is to point out that we can take the amplitude of the energy density variation

$$0 \leq E^2 + B^2 \leq E_{\max}^2 + B_{\max}^2 \quad (5)$$

so we see that a direct wave in the energy density of vacuum is produced. By definition, that is a purely gravitational wave!

Thus scalar electromagnetics is a means of directly converting EM energy to G energy and vice versa.

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Note that ^{the present} replacement of such a system with a zero vector is incorrect. The entity we ~~have~~ ^{call} something EM in vacuum that has energy locked in vacuum, but ~~is not an~~ ^{is not an} E field or B field, is a potential. So the zero-vector in this case should be replaced with a special potential, Φ_s , -- not a 0.

Note that Φ_s has a ^{dynamically} polarized structure, however, instead of a randomized one, in the virtual plasma of vacuum. It thus polarizes the vacuum! Easily and simply.

The part of your paper that electrified me was your showing that these scalar wave effects do affect the Schroedinger equation. For some time I've been pointing this out as vigorously as I can -- and you show it mathematically. Beautiful! We can "pre-engineer" (at least in the future we will be able to) what shall emerge in physical reality as quantum change.

My very best to you, and let's stay in touch. We are at the beginning of a vast new physics.

Sincerely,
Tom B.

P.S.

A quick note: (These systems will be engineerable);

Many of our vector equations may be equated to zero or a zero-vector, with terms A_i such as

$$A_1 + A_2 + \dots + A_i + \dots + A_n = \bar{0}$$

Please notice the infinite variety of things that can be done to this equation when the $\bar{0}$ is regarded as a "special potential" and a "boundary". We may substitute for the $\bar{0}$ any number of special potentials $\bar{\Phi}_s$ which have unique, dynamic substructure that is, to show an example,

$$A_1 + A_2 + \dots + A_i + \dots + A_n = \bar{\Phi}_s = \bar{\alpha}_1 + \bar{\alpha}_2$$

where α_1 and α_2 are dynamic time varying vectors, or even vector systems. At any level, another vector zero system can be ~~be~~ included, such as

$$\bar{\alpha}_1 = \alpha_1 + \bar{0} = \alpha_1 + \bar{\Phi} \quad \text{etc}$$